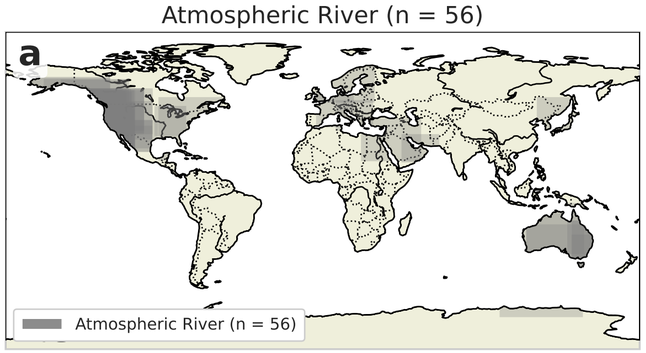
Atmospheric River (n = 56)

a



b

Critical Success Index

Spatial Displacement

Early Signal

	1	3	5	7	10
IFS HRES	0.28	0.21	0.13	0.08	0.04
GraphCast	0.65	0.49	0.37	0.21	0.09
Pangu	0.65	0.48	0.32	0.16	0.06
AIFS	0.63	0.49	0.37	0.22	0.11

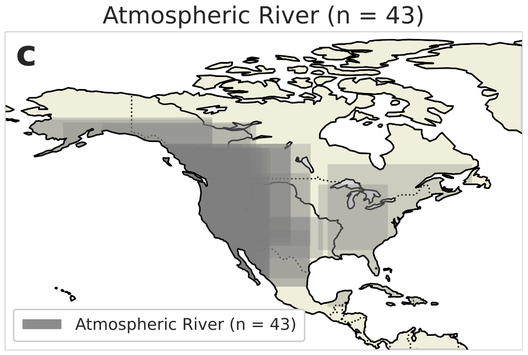
	1	3	5	7	10
IFS HRES	99	167	290	432	560
GraphCast	93	145	231	336	555
Pangu	91	154	226	360	588
AIFS	93	131	212	345	559

	1	3	5	7	10
IFS HRES	0.93	0.85	0.91	0.80	0.67
GraphCast	0.95	0.93	0.93	0.89	0.70
Pangu	0.95	0.95	0.93	0.84	0.80
AIFS	0.95	0.91	0.89	0.84	0.82

Lead Time [Days]

Atmospheric River (n = 43)

c



d

	1	3	5	7	10
IFS HRES	0.26	0.18	0.12	0.07	0.03
GraphCast	0.64	0.48	0.36	0.19	0.08
Pangu	0.64	0.45	0.30	0.13	0.05
AIFS	0.60	0.46	0.34	0.18	0.09

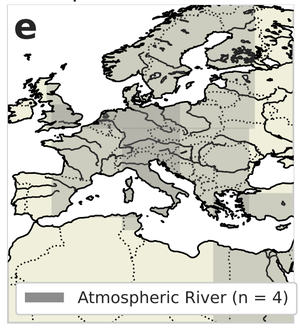
	1	3	5	7	10
IFS HRES	101	174	307	453	553
GraphCast	88	145	223	344	548
Pangu	85	142	218	354	607
AIFS	93	131	204	352	553

	1	3	5	7	10
IFS HRES	0.90	0.81	0.88	0.79	0.69
GraphCast	0.94	0.91	0.91	0.84	0.66
Pangu	0.94	0.94	0.94	0.81	0.75
AIFS	0.94	0.91	0.88	0.81	0.78

Lead Time [Days]

Atmospheric River (n = 4)

e



f

	1	3	5	7	10
IFS HRES	0.33	0.24	0.19	0.12	0.02
GraphCast	0.57	0.44	0.45	0.26	0.02
Pangu	0.55	0.47	0.31	0.13	0.02
AIFS	0.55	0.38	0.32	0.09	0.03

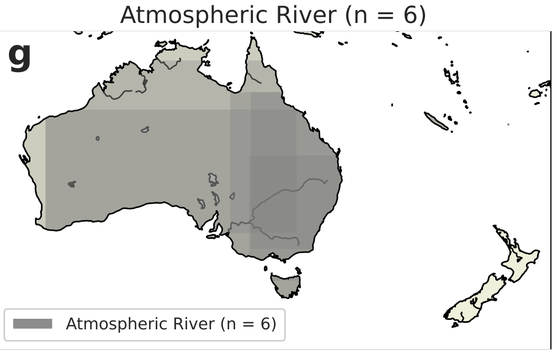
	1	3	5	7	10
IFS HRES	84	127	155	295	638
GraphCast	83	137	195	287	589
Pangu	96	207	230	336	602
AIFS	72	133	218	385	818

	1	3	5	7	10
IFS HRES	1.0	1.0	1.0	1.0	0.5
GraphCast	1.0	1.0	1.0	1.0	0.7
Pangu	1.0	1.0	0.7	0.7	1.0
AIFS	1.0	0.7	0.7	0.7	0.7

Lead Time [Days]

Atmospheric River (n = 6)

g



h

	1	3	5	7	10
IFS HRES	0.36	0.33	0.17	0.12	0.09
GraphCast	0.69	0.53	0.39	0.28	0.18
Pangu	0.70	0.52	0.44	0.29	0.12
AIFS	0.71	0.57	0.43	0.33	0.20

	1	3	5	7	10
IFS HRES	96	146	253	521	570
GraphCast	104	138	267	373	628
Pangu	107	182	228	375	577
AIFS	107	124	240	348	515

	1	3	5	7	10
IFS HRES	1.0	1.0	1.0	0.7	0.7
GraphCast	1.0	1.0	1.0	1.0	1.0
Pangu	1.0	1.0	1.0	1.0	1.0
AIFS	1.0	1.0	1.0	1.0	1.0

Lead Time [Days]

% difference vs IFS HRES

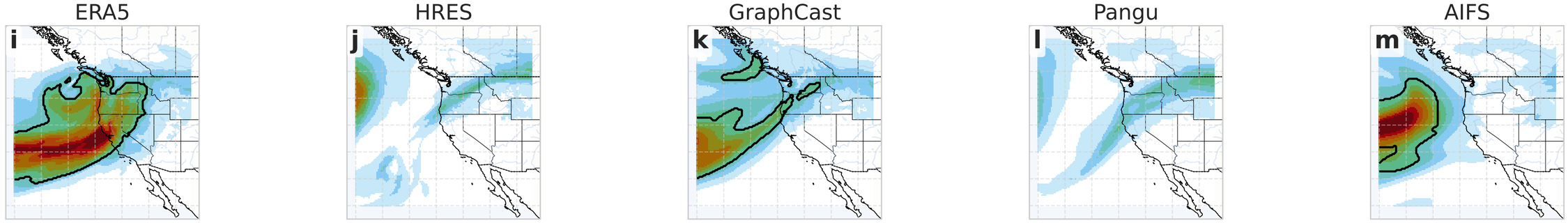
← Worse

Better →

50 20 10 5 2 1 −1 −2 −5 −10 −20 −50

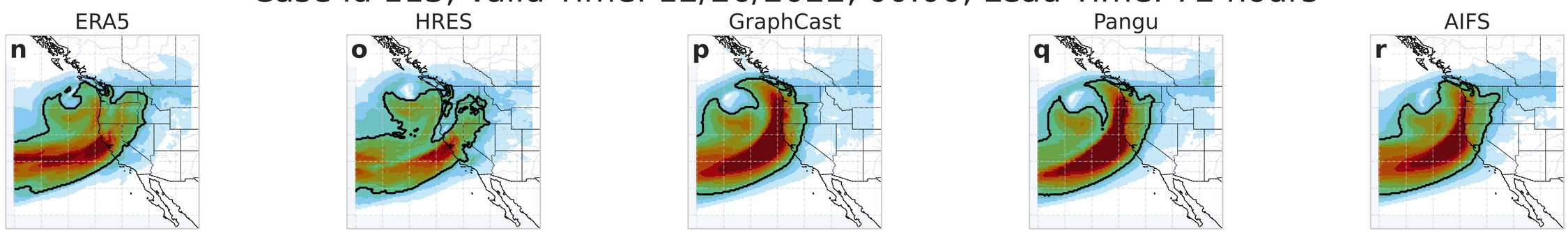
Case id 113, Valid Time: 12/26/2022, 00:00, Lead Time: 240 hours

ERA5 HRES GraphCast Pangu AIFS



Case id 113, Valid Time: 12/26/2022, 00:00, Lead Time: 72 hours

ERA5 HRES GraphCast Pangu AIFS



0 200 400 600 800 1000

Integrated Vapor Transport (kg m^{-1} s^{-1})